

Term Project Guideline

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One of important parts of this course is the class project, which gives students a chance to apply deep neural networks to research oriented topics or real-world applications. This was written based on the project guideline of [CS230](#).

Project Topics

I would like you to choose a meaningful project topic that fits your and/or your team's interests. The topics are usually one of two kinds.

- **Application project:** This is by far the most common. Pick an application that interests you, and explore how best to apply deep learning to solve it.
- **Algorithmic project:** Pick a problem or family of problems, and develop a new learning algorithm, or a novel variant of an existing algorithm, to solve it.

Feel free to propose ambitious things that you are excited about. Just be sure to ask me or TA for help if you are uncertain how to get started. If you are already working on a research or industry project that deep learning might apply to, then you may already have a great project idea.

Previously, students had worked on a wide range of topics. For example,

- Normal map generation with pix2pix model
- Malware classification with convolutional neural networks
- Building Korean food image classifier
- Detection and diagnosis of dental caries with Tensorflow Object Detection API
- Nondestructive inspection using deep learning
- Deep neural networks for peer-to-peer lending

These topics are just examples. I expect you to find more challenging and interesting topics. If you feel hard to find a fancy topic, joining Kaggle competition (it should be challenging) may be a great way to meet the objective of this project.

Project Hints

Once you have identified a topic of interest, it can be useful to look up existing research on relevant topics by searching related keywords on an academic search engine such as: <https://scholar.google.co.kr>. Another important aspect of designing your project is to identify one or several datasets suitable for your topic of interest. If that data needs considerable pre-processing to suit your task, or that you intend to collect the needed data yourself, keep in mind that this is only one part of the expected project work, but can often take considerable time. I expect a solid methodology and discussion of results, so pace your project accordingly.

Two key components should be ready to get started your project.

- Computational resources, especially GPU: Use Colab service as usual. If you need more resources, you can use Google Cloud Platform. It offers \$300 free credit if you are a new customer. Please refer to [this article](#) to learn how to set up Google Cloud instance.
- Preprocessed datasets: The process of inspecting and visualizing the data, trying out different types of preprocessing, and doing error analysis is often an important part of machine learning. Hence if you choose to use preprepared datasets (e.g. from Kaggle, the UCI machine learning repository, etc.), I encourage you to do some data exploration and analysis to get familiar with the problem.

AI Tool Usage Policy

You are allowed to use AI-based tools (e.g., ChatGPT, Claude, GitHub Copilot, etc.) to assist your project work, including coding, debugging, literature search, and writing. However, the following rules must be strictly observed:

- Disclosure requirement: If you use any AI tool at any stage of the project, you must clearly state (1) which tool you used, (2) for what purpose, and (3)

in which part of the project it was applied. This applies to both the proposal and the final presentation materials.

- Where to disclose: Include an "AI Tool Usage Statement" section at the end of your proposal and final submission. For example: "We used ChatGPT (GPT-4o) for debugging data preprocessing code in Section 3.2, and Claude for proofreading the proposal draft."
- Your responsibility: You are fully responsible for the correctness and originality of all submitted content. AI-generated outputs must be critically reviewed and verified by you. Simply copying AI outputs without understanding or verification is not acceptable.
- Undisclosed usage: Failure to disclose AI tool usage will be treated as an academic integrity violation.

Project Deliverables

Proposal



Deadline: April 30, Thursday 14:00

Submission Format: A PDF document (maximum 4 pages)

Presentation: None

Weighting: 10% of total grade

In the project proposal, you will receive feedback from me and TA for your topic. Note that there is a page limit, so I encourage you to **write and organize the contents very clearly** so that I and TA can catch your points without any ambiguity. The following information should be included in your proposal:

- What is the problem that you will be investigating? Why is it interesting? Why do you choose this topic?
- What dataset are you using? If needed, how do you plan to collect it? Please describe the dataset as much as you can.
- What method or algorithm are you proposing? If there are existing implementations, will you use them and how? How do you plan to improve or modify such implementations?

- What reading do you examine to provide context and background? What papers (previous works) do you refer to?
- How will you evaluate your results? Qualitatively, what kind of results do you expect (e.g., plots or figures)? Quantitatively, what kind of analysis will you use to evaluate and/or compare your results? (e.g., what performance metrics)?

Please refer to the table below for the detailed evaluation criteria.

Evaluation Criteria

Aα Criteria	≡ Sub-criteria	# Points
<u>Topic</u>	Is the problem clearly presented?	10
<u>Untitled</u>	Is the motivation convincingly described?	10
<u>Dataset</u>	Are the datasets appropriate to solve the problem?	10
<u>Untitled</u>	Does the team understand well about the datasets?	10
<u>Approach</u>	Is the team's approach suitable for the problem?	10
<u>Untitled</u>	If applicable, did the team search and present existing implementations?	10
<u>Untitled</u>	Did the team review previous works enough?	10
<u>Outcome</u>	Is the evaluation procedure rigorous?	10
<u>Untitled</u>	Do they present the expected results in detail?	10
<u>Writing</u>	Is the proposal written with high clarity?	10

Final Presentation



Deadline: June 4, Thursday 14:00

Submission Format: Presentation material + source code = **a single Colab file**

Presentation: 12 minutes presentation with the submitted materials

Weighting: 20% of total grade

The final material should contain a comprehensive account of your project. I expect your material to be thorough, yet concise. I will be looking for the following contents:

- Good motivation for the project and an explanation of the problem statement
- A description of the data
- Any hyperparameter and architecture choices that were explored
- Presentation of results
- Analysis of results
- Any insights and discussions relevant to the project
- References
- (Extra credit) Member's contribution statement
 - Please describe each member's contribution in detail. Blind peer review will be conducted after the final presentation.
- (Extra credit) Debugging experience worth sharing
- (Extra credit) The well-organized Github repository with the commit history

I would highly recommend submitting a Colab file. With a Colab file, you can write some text, codes and put even figures. The presentation with this file is very welcome. If you plan to use Google Cloud, you can use Jupyter notebook as an alternative (very similar to Colab environment).

Below shows the detailed grading criteria for your project. 10 points are allocated for every sub-criteria except Extra Credit. Sub-criteria in Extra Credit has 2 points each.

Evaluation Criteria

Aa Criteria	≡ Sub-criteria	# Points
<u>Completeness</u>	How challenging is the topic?	10
<u>Untitled</u>	Quality of implementation	10
<u>Untitled</u>	Are the arguments strong enough?	10
<u>Untitled</u>	The process of drawing conclusions	10
<u>Modeling Process</u>	Clear description of data gathering and preprocessing process	10
<u>Untitled</u>	Is the code implemented correctly?	10
<u>Untitled</u>	Hyperparameter selection process	10
<u>Untitled</u>	The detailed analysis of experimental results	10
<u>Presentation</u>	Communication	10
<u>Untitled</u>	Materials	10
<u>Extra Credit</u>	Detailed contribution statement	2
<u>Untitled</u>	Debugging experience	2
<u>Untitled</u>	Github repository	2